

Mechanics Applied to Aerospace Engineering

LABORATORY GUIDELINES

Version 2026

AEROSPACE ENGINEERING DEPARTMENT (UC3M)

Preface

This document contains the guidelines and the work description for the Laboratory and Computer sessions of the course “Mechanics applied to Aerospace Engineering.”

Many of the current and former staff of the Aerospace Engineering department are or have been involved at some point in this course, and have contributed to this document.

The latest version of this document can be found on [Aulaglobal](#). If you detect any errata, or have any comments, please contact Alberto Marín (almarinc@pa.uc3m.es).

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General guidelines

The aim of these Laboratory and Computer sessions is to familiarize the student with (i) the computation of numerical solutions of mechanical problems, (ii) the operation and measurement of mechanical experiments in the laboratory, and (iii) the writing of scientific technical reports. A total of 4 sessions will take place in this course:

1. Particle connected to a spool,
2. Particle on oscillating loop,
3. Compound double pendulum (experimental testing),
4. Compound double pendulum (numerical integration).

Work and evaluation procedure

The development of the Laboratory and Computer sessions will follow the guidelines below:

1. Work shall be done by groups of three students. These are to be established at the first session and **will remain fixed** for all the subsequent sessions.
2. The deliverable of each session consists of a report and the Matlab code used to generate the reported results.
3. All files will be submitted via Aulaglobal in the corresponding Laboratory group (LA, or LB) by the deadline, by only one of the students in each group. All files must be named as indicated in the documentation provided for each session.
4. Failure to follow the guidelines on report writing below will result in a zero (0) grade.

Report writing

Technical reports are used by scientists and engineers to communicate results. The ability to write clear, concise reports is an essential competence to almost any professional. In technical reports, plots and descriptions must follow some quality standards. This document shows basic aspects that must be followed to generate clear plots and express ideas properly in the Laboratory and Computer session reports.

Correct Plotting

Graphs are an essential tool to show results of all types and care should be given to present them in the best way.

Units in graphics: Graphs must have *magnitude* and *units* in their axes, unless they are non-dimensional quantities with no units (this should be indicated using the symbol [-]). In Fig. 0.1 the left and central plots show incorrect examples. The right plot in Figure 0.1 is a correct example. Also, the font face and size used in the plots must be easily readable. The Matlab code to generate this figure (with three subplots) is given in script 1.

Listing 1: Script producing three subplots, with only one fully correct

```

1 close all, clear all;
2 % Script to produce three plots in subfigure
3 % Create the vector of points for the independent variable
4 x = linspace(0,5,100);
5
6 % Compute the dependent variable
7 y = sin(x);
8
```

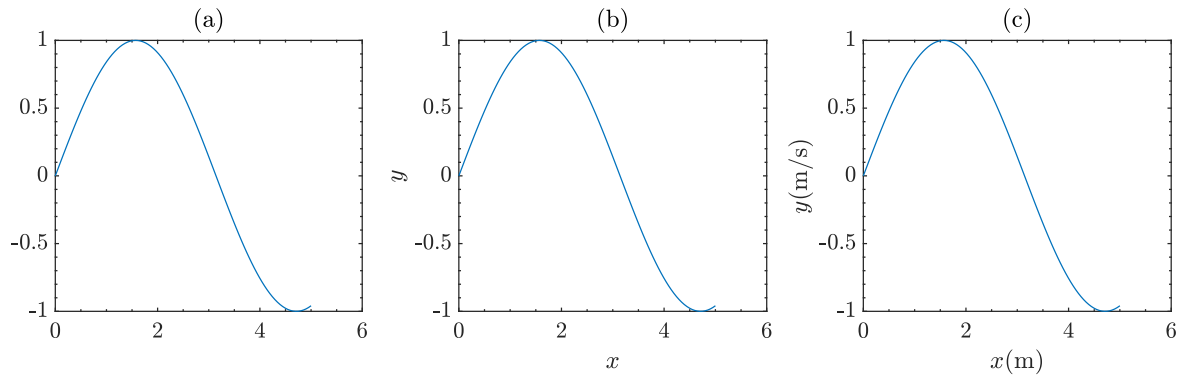


Figure 0.1: Representation of a variable y as a function of the variable x . Subplots (a) and (b) are incorrect examples; (c) is a valid one.

```

9  font_size = 18;
10 figure();
11 % Generate the first subplot in position 1, of a grid with 1 line and 3
12 % columns (1,3)
13 subplot(1,3,1);
14 plot(x,y);
15 title('(a)');
16 set(gca, 'fontsize', font_size);
17
18 % Generate the second subplot in position 2, of a grid with 1 line and 3
19 % columns (1,3)
20 subplot(1,3,2);
21 plot(x,y);
22 title('(b)');
23 xlabel('$x$');
24 ylabel('$y$');
25 set(gca, 'fontsize', font_size);
26
27 % Generate the third subplot in position 3, of a grid with 1 line and 3
28 % columns (1,3)
29 subplot(1,3,3);
30 plot(x,y);
31 title('(c)');
32 xlabel('$x$(m)');
33 ylabel('$y$(m/s)');
34 set(gca, 'fontsize', font_size);
35
36 % Save the figure
37 set(gcf, 'Units', 'Normalized', 'OuterPosition', [0, 0.04, 0.8, 0.45]);
38 saveas(gcf, '../figs/figure_1.eps', 'eps')

```

Multiple curves in the same graph: Combining widely different magnitudes on the same axes must be avoided, as it could lead to errors in the understanding of the results.

For example, consider that two magnitudes (y_1 and y_2) are to be plotted in the same graph in the same interval of x . If the values of one of them are much greater than the values of the other,

the smallest one will be barely visible. Take, for instance:

$$y_1 = x^2 \quad (0.1)$$

$$y_2 = -100000000x^3 \quad (0.2)$$

If both magnitudes are plotted on the same axis, as it is done with script 2, the results for y_1 are not clear. Figure 0.2 shows the graph obtained with script 2.

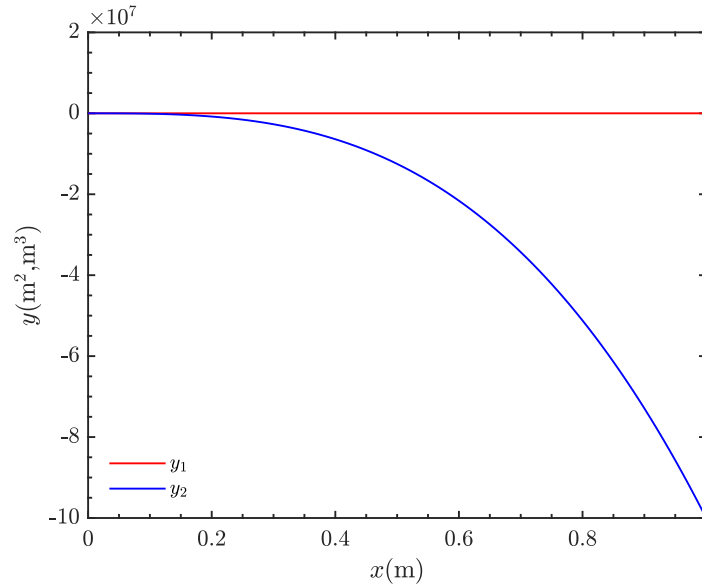


Figure 0.2: Wrong

Listing 2: Inadequate plotting for variables differing by several orders of magnitude

```

1 close all , clear all
2
3 % Define x, y1 and y2
4 x = 0:0.01:1;
5 y1 = x.^2 ;
6 y2 = -100000000*x.^3 ;
7
8 % Plot y1
9 plot(x, y1, 'r', x, y2, 'b')
10
11 % Set the label for the two axes + the legend
12 xlabel('x(m)');
13 ylabel('y(m^2, m^3)');
14 h = legend('y_1$', 'y_2$', 'Location', 'SouthWest');
15 set(h);
16
17 saveas(gcf, '../figs/figure_2.eps', 'eps')

```

If we insist on plotting such two variables on the same graph, we can use separate vertical axes for y_1 and y_2 , each with its different scale and units. Then, both magnitudes will be clearly understandable. Script 3 generates a proper graph for two magnitudes in the same axis, while Figure 0.3 shows the corresponding graphs. Also, the limits of the axes should be chosen to display correctly the data to be presented.

A valid alternative in some cases (when the data is properly adimensionalized) is to use logarithmic scales in the plot. Logarithmic plots are also useful when a single variable varies widely in scale (either in the horizontal and vertical axes) and we want to highlight features on those different scales.

It takes some experimentation to create valid and correct plots to transmit clearly what you intend, to your readers.

Listing 3: Adequate plotting of variables differing by several orders of magnitude

```

1  close all , clear all
2
3  % Define x, y1 and y2
4  x = 0:0.01:1;
5  y1 = x.^2;
6  y2 = -100000000*x.^3;
7
8  % Get current axes handle while plotting the two lines (y1 and y2)
9  [ax,h1,h2] = plotyy(x,y1,x,y2);
10 % Set the colors of the plotted lines and axes
11 set(h1, 'Color', 'r');
12 set(h2, 'Color', 'b');
13 set(ax,{ 'ycolor' },{ 'r'; 'b' })
14
15 % Specify labels for the axes
16 xlabel('x(m)')
17 ylabel(ax(1), 'y1(m^2)')
18 ylabel(ax(2), 'y2(m^3)')
19 % Set the legend
20 legend('y1','y2','Location','West')
21 vv = get(ax(2), 'Position');
22
23 set(ax(2), 'Position', [vv(1) vv(2)+0.05 vv(3)*0.9 vv(4)*0.9])
24 saveas(gcf, '../figs/figure_3.eps', 'epsc')

```

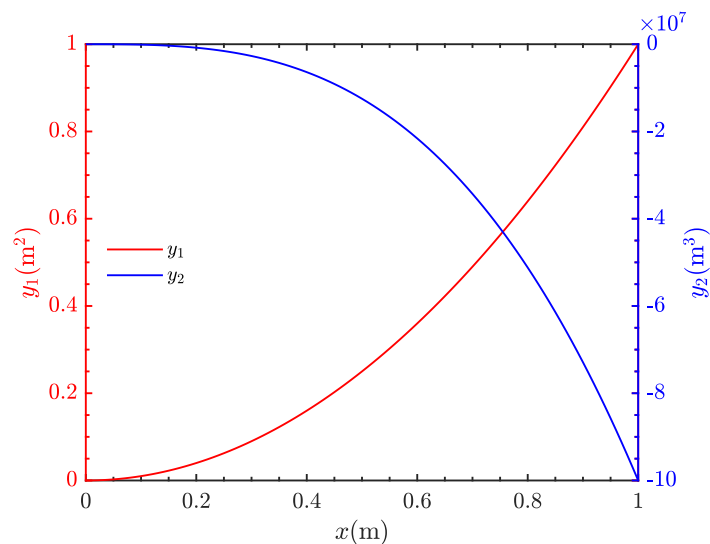


Figure 0.3: Correct way of displaying two variables of widely different magnitude in the same plot.

Sampling data points correctly: When Matlab (or any other programming language) generates a curve, it uses discrete points. The number of points used is important for the graphical aspect of the output. Nyquist's theorem states that a time signal can be correctly interpreted (with Fourier's analysis) if the sampling frequency is at least twice the highest signal frequency (for a sinusoidal signal, this is f_0 , so that $f_s > 2f_0$). This condition enables to correctly reproduce the fundamental frequency

of the signal (i.e. the number of oscillations in a given period), however, for a proper representation of the signal, you should use a higher frequency, as shown in Figure 0.4, which compares the resulting curve of sampling $y = \sin(t)$ ($f_0 = 1/(2\pi)$) with three different frequencies (refer to Script 4 for the required code to generate the figure). Clearly, when below the Nyquist's limit, the fundamental frequency (5 oscillations in the plotted time interval) is not even captured. For a frequency slightly higher than $2f_0$, on the other hand, the fundamental frequency is captured, although the curve still appears very noisy and useless for a proper signal representation. For a sampling frequency of $15f_0$, finally, the curve already represents properly a sine function.

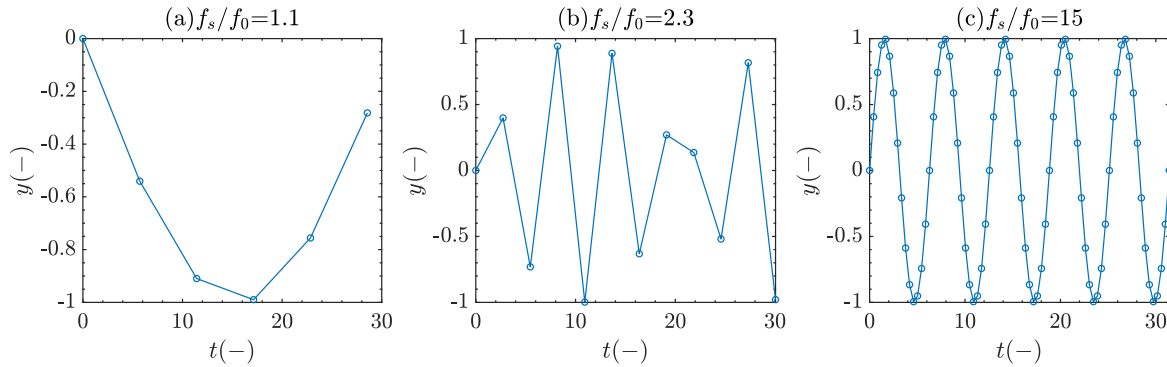


Figure 0.4: Plot of $y = \sin(t)$ (in non-dimensional units) obtained for three different sampling frequencies: (a) $f_s = 1.1f_0$, (b) $f_s = 2.3f_0$, and (c) $f_s = 15f_0$. Observe that non-dimensional units are indicated with $(-)$.

Listing 4: Sampling script to obtain the 3 different plots of $y = \sin(t)$, for 3 different sampling scripts

```

1 clear all; close all;
2 % Frequency of our problem ( y = sin ( 2 pi f * t )
3 f = 1/(2*pi);
4
5 % Sampling frequency below Nyquist limit
6 fs = 1.1*f;
7 % Distance between consecutive x-points
8 step = 1/fs;
9 % Vector t and y
10 t = 0:step:5*(2*pi);
11 y = sin(t);
12
13 font_size = 18;
14 % Plot in the first subplot
15 subplot(1,3,1);
16 plot(t,y,'marker','o');
17 title(' (a) $f_s/f_0=1.1$', 'FontWeight','Normal', 'fontsize', font_size);
18 xlabel('$t(-)$'); ylabel('$y(-)$');
19 set(gca, 'fontsize', font_size);
20
21 % Sampling frequency above Nyquist limit
22 fs = 2.3*f;
23 step = 1/fs;
24 % Vector t and y
25 t = 0:step:5*(2*pi);
26 y = sin(t);
27 % Plot in the second subplot
28 subplot(1,3,2);

```

```
29 plot(t,y,'marker','o');
30 title('(b) $f_s/f_0=2.3$', 'FontWeight','Normal');
31 xlabel('$t(-)$'); ylabel('$y(-)$');
32 set(gca, 'fontsize', font_size);
33
34 % Sampling frequency well above Nyquist limit
35 fs = 15*f;
36 step = 1/fs;
37 % Vector t and y
38 t = 0:step:5*(2*pi);
39 y = sin(t);
40 % Plot in the third subplot
41 subplot(1,3,3);
42 plot(t,y,'marker','o');
43 title('(c) $f_s/f_0=15$', 'FontWeight','Normal', 'fontsize', font_size);
44 xlabel('$t(-)$'); ylabel('$y(-)$');
45 set(gca, 'fontsize', font_size);
46
47 % Save the figure
48 set(gcf, 'Units', 'Normalized', 'OuterPosition', [0, 0.04, 0.8, 0.45]);
49 saveas(gcf, '../figs/figure_sampling_frequency.eps', 'epsc');
```

Confusing point patterns: A final important point to be aware of when plotting continuous curves as discrete points, is the appearance of confusing patterns. These may arise due to purely a numerical artifact, or some kind of resonance between the fundamental frequency of the signal and the sampling frequency. Script 5 generates a figure that may be confusing, which is shown in Fig. 0.5 for a correct sampling rate ($f_s = 8.1f_0 > 2f_0$) of the same function considered before ($y = \sin(t)$), but it is clear that the resulting pattern may confuse the reader, who hardly sees a sinusoidal oscillation. This can be solved by increasing the sampling frequency, or adding (where it makes sense) lines connecting the points, as in Figure 0.4.

Listing 5: Confusing patterns

```
1 close all, clear all
2 % Frequency of our problem
3 f = 1/(2*pi);
4 % Sampling frequency
5 fs = 8.1*f;
6 % Distance between consecutive x-points
7 step = 1/fs;
8 % Vector x and y
9 x = 0:step:5*(2*pi);
10 y = sin(x);
11 % PLOT
12 plot(x,y,'o')
13 title('Confusing_patterns_for_the_function_$y=\sin(t)$');
14 xlabel('$t(-)$'); ylabel('$y(-)$');
15
16 saveas(gcf, '../figs/figure_confusing_patterns.eps', 'epsc')
```

Customizing style and fonts in plots: In your reports, plots should be as much homogeneous as possible to ease their understanding. This can be achieved by modifying the default settings of the graphics root object of the Matlab renderer. This way, you do not need to specify the font size and other properties for each of your figures.

An example of such a script (which should be executed once every time you open Matlab) is given in Script 6.

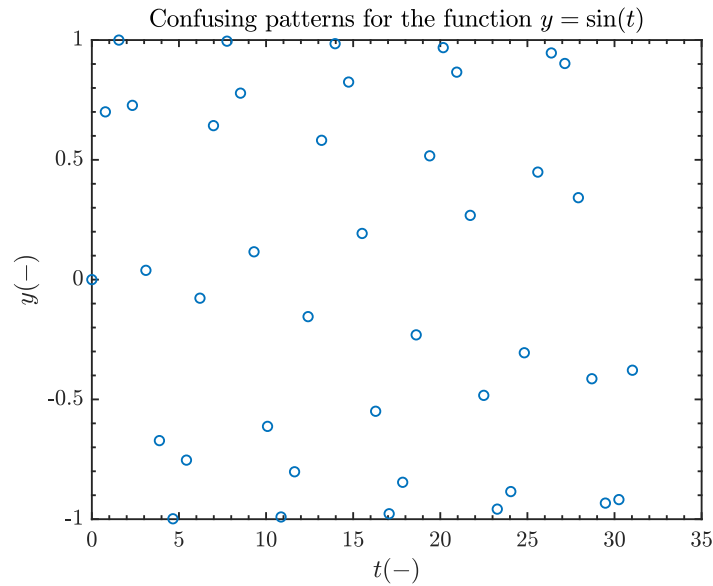


Figure 0.5: Correct but confusing representation of $y = \sin(t)$, with sampling frequency well above Nyquist's limit: $f = 8.1f_0$.

Listing 6: Use of Latex font in a Matlab figure

```

1 %% MODIFY DEFAULT SETTINGS IN MATLAB (run only once at session startup)
2
3 %{
4 The following is a set of commands that can be helpful
5 to maintain a fixed style and format for all your figures.
6 They set some default properties of the graphics root object
7 of the Matlab renderer.
8 Note that these settings are maintained throughout the
9 Matlab session, until it is restarted,
10 so this script needs to be run only once,
11 at the beginning of each session.
12 %}
13
14 % set the size of the ticks on the axes
15 set(groot, 'defaultLegendFontSize', 12);
16 % set the default size of the text
17 set(groot, 'defaultTextFontSize', 12);
18 % set the default axes font size.
19 set(groot, 'defaultAxesFontSize', 12);
20
21 % set the width of the axes
22 set(groot, 'defaultAxesLineWidth', 1);
23 % activate the minor ticks of the axes
24 set(groot, 'defaultAxesXMinorTick', 'on');
25 set(groot, 'defaultAxesYMinorTick', 'on');
26 % deactivate the legend by default
27 set(groot, 'defaultLegendBox', 'off');
28 % define the default line width in the plots
29 set(groot, 'defaultLineLineWidth', 1);
30 % define the default line marker size
31 set(groot, 'defaultLineMarkerSize', 5);

```

```
32 % set the font of the axes ticks to Latex
33 set(groot, 'defaultAxesTickLabelInterpreter', 'latex');
34 % define the font for the default text for the rest of
35 % objects (labels, titles, legend, etc...)
36 set(groot, 'defaultTextInterpreter', 'latex');
37 set(groot, 'defaultLegendInterpreter', 'latex');
```

Then, you have two possibilities for generating homogeneous plots for your report:

1. you either run the Script 6 every time you open Matlab and then you avoid specifying the figure properties, unless you have a very specific need, or
2. you specify the figure properties always and in the same way within your figure code, possibly using a dedicated plotting function.

These two approaches are shown in Script 7, while the resulting figure (which is exactly the same, if you run the default settings script before) is shown below in Figure 0.6. You will have noticed that, the previous figures were generated by running the default settings script, and therefore they feature the Latex font and the rest of properties defined there.

Listing 7: Two different approaches to generate the same Matlab figure

```
1 % delete all pre-existing contents of the workspace
2 clear all
3 % clos all open figures/objects
4 close all
5 % delete all text in the command window
6 clc;
7
8 % Example data to plot
9 t = 0.0:0.1:5;
10 y = exp(-t);
11
12 % Plot the data
13 figure;
14 plot(t,y, '-o');
15 % Plot title and set the Latex interpreter
16 title('Exponential_decay_<math>y=\exp(-t)</math>', 'fontsize', 12)
17 l = legend('<math>\exp(-t)</math>');
18 set(l, 'fontsize', 12);
19 % Plot the axes labels with Latex interpreter
20 ylabel('<math>y</math>(m)', 'fontsize', 12)
21 xlabel('<math>t</math>(s)', 'fontsize', 12)
22
23 % Save the figure
24 saveas(gcf, '../figs/latex_font_figure.eps', 'epsc');
```

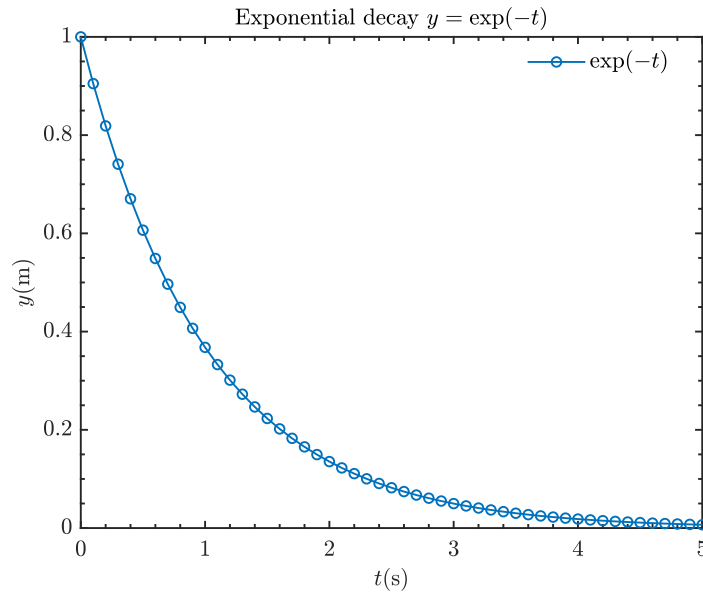


Figure 0.6: Plot of an exponential decay, written with the Latex interpreter for title, labels and axes ticks.

Technical report writing tips

Technical writing is an essential competence that all engineers must possess. Some important considerations must be taken into account when writing the Laboratory and Computer session reports:

- A technical report has to be concise and precise. Brief is preferred to verbose, and precision is preferred to vagueness.
- Begin any technical writing by first sketching the sections and then the paragraphs that will go in each section.
- Do not express ideas in first singular person (first plural person is sometimes fine, although impersonal forms are preferred).
- Do not use language contractions or slang words. Avoid clichés and overused stock phrases: they sound good, but provide no information. Avoid ambiguous adverbs like “very”, “much”, etc. if they provide no information.
- Technical reports should follow a clear structure. A good example is to include the following sections: Introduction, Methodology (including the mathematical model/experimental setup), Results, Discussion, Conclusions. Usually, a short abstract at the beginning of the document summarizes the report.
- Commonly, a brief summary/abstract is included at the top of the report, which must be self-contained. Anyone reading the abstract should be able to grasp what the paper is about, its main hypotheses, methods, results, and conclusions.
- The introduction section must explain what is the purpose of this report and prepare the reader for what he/she will find in it. The description of the problem must detail the subject of the report, like a physical problem or phenomenon.
- The methodology section describes how the problem is tackled, by means of analytical models, numerical code, simulations, experiments, etc. The different parts of the model, its assumptions, equations, the numerical schemes to be used, etc. are contained in this section. In the case of experimental work, the experimental setup to be used, the different instruments, etc. are described in detail in this section.
- In the results section, describe the facts and results. Do not write vague thoughts, suppositions, or anything that is not a clear fact in this section of the report. Regarding the discussion of results: You can either interpret and discuss the results as you present them, or gather all comments and interpretations for a dedicated discussion section.

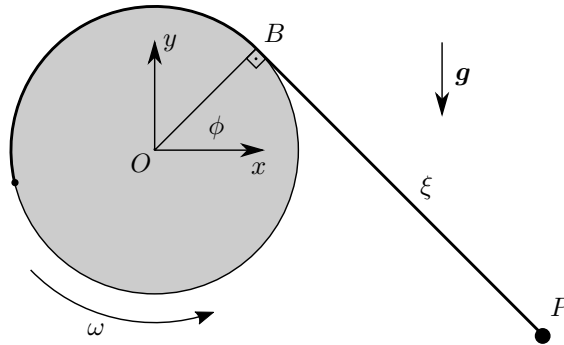
- The conclusions section must gather the main results/conclusions of the work.
- Try to maintain a given format/style and font for the full report, including the text in written paragraphs, but also in tables and figures. Each variable should have a single symbol in your report.

Additional advice

Use numbered captions in figures and tables, and refer to these figures in the text. A final *bibliography* section should list any papers or books referenced in the work, which in turn must be cited from the main text. Most text editors allow the user to set captions and labels in an easy way. Look up for labels and cross-references if you use MS Word. You may also be interested in using L^AT_EX (e.g. on [Overleaf](#)), ideal for technical reports with many references to figures, citations, and mathematics.

1 Particle connected to a spool

A heavy particle P of mass m is connected by a string of total length l to a cylindrical spool of radius a and center at O as shown in the figure. The spool rotates at a known constant angular rate ω counterclockwise. The string is massless, inelastic, thin, and remains always in tension during the motion of the particle. We define the inertial reference frame Oxy as shown in the figure. The other end of the string is attached to a point on the spool, and we shall call B the instantaneous point of tangency between the straight segment of the string (the segment BP) and the spool. We will denote the angle between the local horizontal (axis Ox) and the radius OB as ϕ .



The purpose of this first computer laboratory session is to get used to integrating the equations of motion of a problem in mechanics numerically, producing results, and documenting/discussing them in a technical report.

While this and other problems are not analytically integrable in general, they can be solved numerically. This requires first expressing the equations of motion in a form that a numerical software like Matlab can handle, and then using integration routines (such as `ode45`) to obtain the position and the velocity of the point-mass as a function of time.

1.1 Laboratory session 1

1. Verify that the system of study has only one configuration degree of freedom, that we can parametrize with ϕ . In general, the configuration also depends explicitly on time t because of the rotation of the spool.
2. Express ξ as a function of time t and angle ϕ . Write down the position, velocity and acceleration vectors of P with respect to Oxy as a function of the constants of the problem, time t , and the angle ϕ and its time derivatives.
3. Write down the mathematical expression of the forces acting on P .
4. Write down a second-order differential equation for ϕ that does not depend on the tension in the string. And write an equation for the tension T as a function of ϕ , $\dot{\phi}$.
5. Expand the differential equation for ϕ as two first-order differential equations for the state vector $X = [\phi, \dot{\phi}]^T$:

$$\frac{d}{dt}X = f(t, X) \quad (1.1)$$

Implement this as a Matlab function `Xdot = diffeq(t,X,g,omega,...)`, adding any other parameters of the problem that you need at the end of the function definition. This function will be integrated with `ode45` below.

6. Create a stop function that will detect when the particle hits the spool ($\xi = 0$) or when the tension in the string vanishes ($T = 0$), with the signature `[value,terminate,direction] = stopfun(t,X)`, where `value` is a vector equal to $[\xi, T]$, `terminate` is equal to $[1, 1]$ to stop the integration, and `direction` is equal to $[-1, -1]$.

7. Integrate the differential equations with `ode45` for the study cases described below, passing the function `stopfun` within the options of `odeset`. Browse the help page of `ode45` and `odeset` to learn about these built-in function and get familiar with the Matlab help system. Plot the evolution of ϕ and the amount of unspooled string ξ as a function of time. Plot the tension in the string T as a function of time.
8. Plot the trajectory of P in the Oxy plane in each case. Please add a circle to represent the spool, for reference.¹
9. Write down the expression of the mechanical energy E of P . Plot it as a function of time for each case. Explain in which case(s) it is conserved and why.
10. When and where does the particle hit the spool in case 3? Discuss what happens to $\dot{\phi}$ near the end in case 3. When does the string lose tension in case 4? What is the stop condition in case 5?
11. Open discussion: discuss what changes would need to be made to the code to integrate the motion of the particle after tension is lost in case 4.

Parameters

- $g = 9.81 \text{ m/s}^2$.
- $m = 0.1 \text{ kg}$.
- $a = 20 \text{ cm}$.
- $\phi(0) = 0 \text{ rad}$.
- $\xi(0) = 1 \text{ m}$.

Case 1

- $\omega = 0.1 \text{ rad/s}$.
- $\dot{x}^P(0) = 0 \text{ m/s}$.

Case 2

- $\omega = 0 \text{ rad/s}$.
- $\dot{x}^P(0) = -1 \text{ m/s}$.

Case 3

- $\omega = 0 \text{ rad/s}$.
- $\dot{x}^P(0) = -10 \text{ m/s}$.

Case 4

- $\omega = 0 \text{ rad/s}$.
- $\dot{x}^P(0) = 4 \text{ m/s}$.

Case 5

- $\omega = 0.1 \text{ rad/s}$.
- $\dot{x}^P(0) = 1 \text{ m/s}$.

¹The adventurous students may want to create a video animation of the motion of the particle in time, adding the unspooled segment of the string to the image (see `getframe`).

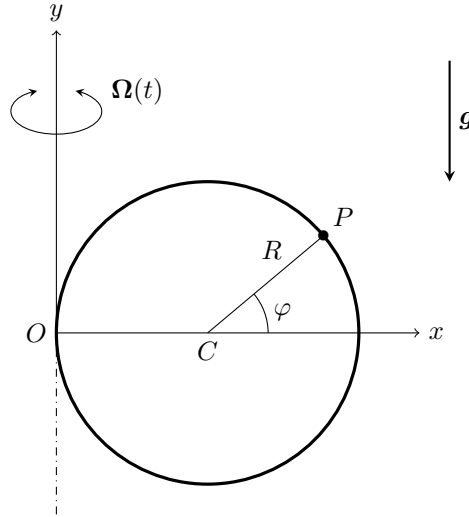
Deliverables

One student in each team will upload all the results to AulaGlobal in a single zipped file named `Group_Li_S1_S2_S3.zip`, where *Li* stands for the Lab Group (LA, or LB) and *S1*, *S2* and *S3* are the first surnames of the three students (e.g., `Group_LA_Sánchez_González_García`). The zipped file must contain the following:

1. A report in pdf format and **maximum 10 pages** gathering the answers to the points above and any other results of the work. Critical analysis of the results is mandatory. Always include in the title page the full names of the three students. The name of the report file must be: `Report_Li_S1_S2_S3.pdf`
2. Code developed to obtain the results, contained in a sub-folder named `code_Li_S1_S2_S3`. The main script must be called `main.m` and must produce all results when run, without any input from the user. The code will be checked by the instructor by running it directly from this folder, so make sure it runs properly before submitting (a penalty will be applied if code does not run or fails to produce the results).

2 Particle on oscillating loop

A heavy point particle P of mass m slides without friction on a circular wire of center C and radius R . The wire is located on a vertical plane Oxy that rotates about the vertical axis Oy with a prescribed angular frequency $\Omega(t) = A\omega \sin(\omega t)\mathbf{j}$, where A and ω are known positive constants, as shown in the figure. The angle that the vector $\mathbf{CP}$ makes with the Ox axis is denoted φ .



2.1 Laboratory session 2

1. Write down the position, velocity and acceleration vectors of P relative to the reference frame $Oxyz$, as a function of φ and its derivatives.
2. Identify the forces that act on P , and write down their mathematical expression for a generic instant of time t .
3. Performing the analysis of the dynamic in the rotating reference frame, write down a second order differential equation for φ that does not depend on the reaction forces. Then, write down the equations for the components of the reaction forces, which may depend on φ and $\dot{\varphi}$.
4. Expand the differential equation for ϕ as two first-order differential equations for the state vector $X = [\varphi, \dot{\varphi}]^T$:

$$\frac{d}{dt}X = f(t, X) \quad (2.1)$$

Implement this as a Matlab function `Xdot = diffeq(t,X,g,omega,...)`, adding any other parameters of the problem that you need at the end of the function definition. This function will be integrated with `ode45` below.

5. Integrate the differential equation with `ode45` considering the following parameters:

- $A = \pi$.
- $g = 9.81 \text{ m/s}^2$.
- $m = 1 \text{ kg}$.
- $R = 1 \text{ m}$.
- $\dot{\varphi}(0) = 0 \text{ rad/s}$.
- $\varphi(0) = 0 \text{ rad}$.
- $\omega = 0.1 \text{ rad/s}$.

for $t = [0, 3T]$ s, where T is the time period of the rotating reference system. Plot the evolution of φ , the reaction forces, the inertia forces, and the kinetic energy and gravitational potential energy as a function of time.

6. Plot the trajectory of P as seen from a stationary reference frame $Ox_1y_1z_1$.²
7. Considering $A = 0$, plot in one figure $\dot{\varphi}$ as a function of φ for all possible combinations of the following initial conditions:

- $\varphi(0) = [0, \pi/4, \pi/2]$ rad
- $\dot{\varphi}(0) = [0, 1.5, 3]$ rad/s

Discuss the results. What are the particular combinations of $\varphi(0)$ and $\dot{\varphi}(0)$ that lead to a value of φ that increases steadily over time? Derive a general analytical expression of $\dot{\varphi}(0)$ as a function of $\varphi(0)$ that will allow to determine whether φ increase over time or oscillates between two values (Hint: use the potential energy and the total mechanical energy of the system).

8. Assume now that $A^2 \ll g/(R\omega^2)$ and that the motion of P is such that $\varphi = \delta - \pi/2$ with $\delta \ll 1$. Integrate the linearized equation using `ode45` for the following values of ω :

- $\omega = \sqrt{g/R}$ rad/s.
- $\omega = (1/2)\sqrt{g/R}$ rad/s.
- $\omega = (1/4)\sqrt{g/R}$ rad/s.

Plot in a single graph the percentage error between the linearized solutions and the solutions of the full differential equation as a function of time (Consider $A = 0.1$ and $R = 0.01$ m).

9. Critically discuss all the results.

Deliverables

One student in each team will upload all the results to AulaGlobal in a single zipped file named `Group_Li_S1_S2_S3.zip`, where Li stands for the Lab Group (LA, or LB) and $S1$, $S2$ and $S3$ are the first surnames of the three students (e.g., `Group_LA_Sánchez_González_García`). The zipped file must contain the following:

1. A report in pdf format and **maximum 10 pages** gathering the answers to the points above and any other results of the work. Critical analysis of the results is mandatory. Always include in the title page the full names of the three students. The name of the report file must be: `Report_Li_S1_S2_S3.pdf`
2. Code developed to obtain the results, contained in a sub-folder named `code_Li_S1_S2_S3`. The main script must be called `main.m` and must produce all results when run, without any input from the user. The code will be checked by the instructor by running it directly from this folder, so make sure it runs properly before submitting (a penalty will be applied if code does not run or fails to produce the results).

²The adventurous students may want to create a video animation of the motion of the particle in time (see `getframe`).

3 Compound Double Pendulum

The compound double pendulum is a classical mechanics system that exhibits rich and complex dynamics. Unlike the idealized double pendulum, where each limb is modeled as a massless rod with a point mass at its end, the compound double pendulum accounts for the distributed mass of each limb and its associated rotational inertia. This realistic model allows for a more accurate representation of the dynamics, capturing behaviors such as chaotic motion, sensitivity to initial conditions, and energy conservation. In this laboratory, students will explore the compound double pendulum through both experimental and numerical approaches, learning how to model a deterministic chaotic system.

The compound double pendulum consists of two rectangular limbs connected by pivots, with the motion of the upper pendulum influencing the lower pendulum and vice-versa. Each limb is equipped with colored markers (blue, red, and green) to track their motion through digital video, corresponding to the three pivot points. Students will collect data, model the system's dynamics, and validate their numerical simulations against experimental results. Hence, the primary objectives of this laboratory session are to:

- Understand the principles governing the motion of a rigid-body compound double pendulum.
- Derive and analyze the equations of motion for the system, incorporating distributed mass and moments of inertia.
- Process experimental video data to extract pivot positions and estimate initial conditions for numerical modeling.
- Implement and solve the system's equations of motion in MATLAB, ensuring accurate results and system stability.

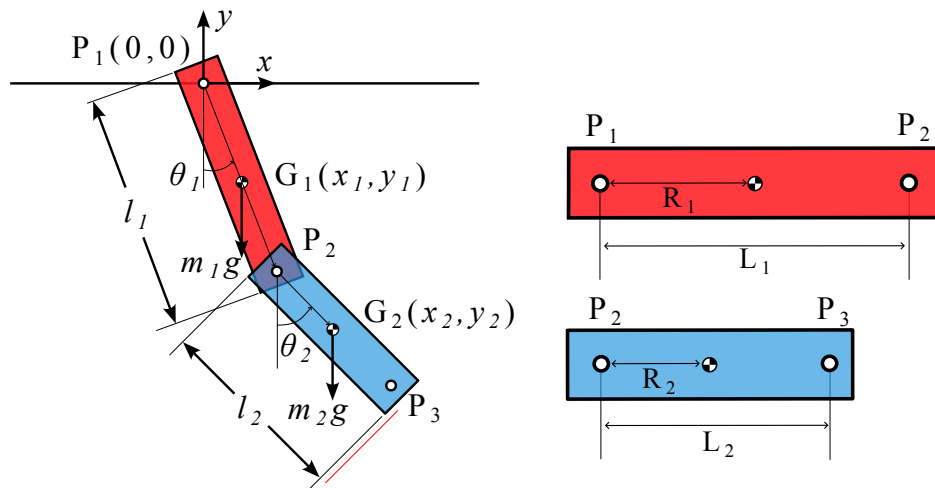


Figure 3.1: Rigid Body Double Pendulum Model

3.1 General description of the problem

Let i be the unit vector along the x axis, and let j be the unit vector along the y axis. We regard y as increasing upwards (this is the usual mathematical convention, as opposed to computer graphics systems which often have y increasing downwards).

The fixed pivot, which we name pivot 1, is at location P_1 . Pivot 2 is the moving pivot point connecting the two pendulums at location P_2 . The third pivot, although useless from the formulation standpoint, is referred to as P_3 . Please refer to figure 3.1. The center of mass of each limb is denoted as G_1 and G_2 respectively.

Let pendulum 1 be the upper pendulum. It is connected to a fixed location at pivot 1 (P_1), and connected to the lower pendulum at pivot 2 (P_2).

- Let $\mathbf{r}_0^{G_1} = x_1\mathbf{i} + y_1\mathbf{j}$ be the location of the center of mass of pendulum 1.

Table 1: Different geometries and masses for the limbs

Length (l)	Pivot distance (L)	Mass (m)	Width (w)	Thickness (t)
350 mm	297 mm	560 ± 1 gr	50 mm	12.5 mm
300 mm	247 mm	480 ± 1 gr		
250 mm	197 mm	400 ± 1 gr		

- Let $\mathbf{P}_1\mathbf{G}_1$ be the vector from pivot 1 to the center of mass of pendulum 1, with length R_1 .
- Let θ_1 be the angle at pivot 1 between $\mathbf{P}_1\mathbf{G}_1$ and the downward vertical position.
- Let $\mathbf{P}_1\mathbf{P}_2$ be the vector from pivot 1 to pivot 2, with length L_1 .

Let pendulum 2 be the lower pendulum. It is connected to the upper pendulum at pivot 2 (P_2).

- Let $\mathbf{r}_0^{G_2} = x_2\mathbf{i} + y_2\mathbf{j}$ be the location of the center of mass of pendulum 2.
- Let $\mathbf{P}_2\mathbf{G}_2$ be the vector from pivot 2 to the center of mass of pendulum 2, with length R_2 .
- Let θ_2 be the angle at pivot 2 between $\mathbf{P}_2\mathbf{G}_2$ and the downward vertical position.

The geometrical and inertial properties of the system are provided by:

- Let l_1 and l_2 be the lengths of the two rectangular bodies.
- Let $w_1 = w_2 = 50\text{mm}$ be the width, and $t_1 = t_2 = 12.5\text{mm}$ the thickness of the limbs.
- Let m_1 and m_2 be the masses of the two bodies.
- Let I_1 and I_2 be the moments of inertia of the two bodies, which can be computed from their dimensions.

3.2 Session 1: Experimental testing

In the first session, students will work in the laboratory to observe and record the motion of a compound double pendulum. This session focuses on acquiring experimental data and preparing it for numerical analysis in the second session. Each group of students will test a unique configuration of the pendulum, varying the limb lengths and masses according to the specifications provided in Table 1. The primary objectives of this session are: (1) to collect experimental data on the motion of the compound double pendulum using a video recording device; (2) to post-process the video data to extract key information about the motion of the pivot points and center of masses; and, (3) to identify the initial conditions of the system, which are essential for the numerical part of the second laboratory session. The main tasks to solve are:

1. **Setup and Recording.** Upon arriving at the laboratory, each group will be assigned a specific configuration of the double pendulum, defined by the masses and lengths of its limbs (see Table 1). Groups will record the motion of the double pendulum using a cell phone or similar device. Ensure the pendulum is visible in its entirety throughout the motion and that the pivot points (blue, red, and green markers) are clearly distinguishable. Launch the double pendulum from an initial condition of your choosing. **Record at least 20 seconds of motion for analysis.** Ensure that the frame rate of the video is sufficient to capture the dynamics of the pendulum (e.g., 60 frames per second).
2. **Initial Condition Identification.** After recording the video, students will analyze the first frame of the video to determine the initial conditions of the experiment. Compute the initial angles θ_1 and θ_2 of the pendulum limbs relative to the vertical axis.
3. **Moment of Inertia.** Compute the moment of inertia of each limb, based on their geometrical and inertial properties, at the centre of gravity ($I_{1,G}$ and $I_{2,G}$). *Note: You must account for the rounded tips of the limbs.*
4. **Post-Processing the Video.** Using MATLAB, students will process the recorded video to extract the positions of the pivot points. Students will be provided with MATLAB code templates for video processing and marker detection in Aulaglobal. The code will track the motion of the blue, red, and green markers to obtain the displacements of the pivot points over time.

Use the displacements of the pivot points to compute the angular positions θ_1 and θ_2 for the double pendulum. Additionally, determine the positions of the center of masses for both pendulum limbs based on their respective geometries. *Note: advanced students may want to refine the video tracking code by applying filtering techniques to remove noise from the extracted data and ensure that the motion of the pivot points and center of masses is accurately captured.*

5. **Discussion:** Use your dataset to estimate the velocity of the center of mass of each limb. Compute the kinetic, potential, and total mechanical energy of the system. Is the total mechanical energy of the system conserved?

3.3 Session 2: Numerical Integration

In the second session, students will implement a numerical solution to model the dynamics of the compound double pendulum using MATLAB. Leveraging the experimental data and initial conditions obtained in the first session, the students will simulate the motion of the pendulum and analyze its dynamics. This session emphasizes coding, numerical integration, energy analysis, and validation against experimental data. The primary objectives of this session are: (1) To implement a MATLAB script for solving the equations of motion of the compound double pendulum using numerical methods; (2) to analyze the motion of the pendulum, extracting angular displacements, pivot positions, center of mass trajectories, and energy values; and, (3) to validate the numerical results by comparing them against the experimental data collected in the first session. An extra activity is proposed to explore extensions, such as introducing damping torques and fine-tuning the model for better agreement with the experimental results. The tasks to be solved are:

1. **MATLAB Implementation.** Using the provided equations of motion (included in this document), students will set up a MATLAB function to calculate the angular accelerations $\ddot{\theta}_1$ and $\ddot{\theta}_2$ as functions of the current state vector $(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$. Ensure the equations account for all forces and moments acting on the system, including the moments of inertia and gravitational forces. Use MATLAB's stiff solver `ode23s` to numerically integrate the equations of motion. This solver is chosen to handle the stiff nature of the system, ensuring stability and accurate energy conservation. Input the initial conditions derived from the experimental session $(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$.
2. **Interpretation of results.** Perform an in-depth analysis of the results, including the evolution of the angular displacements θ_1 and θ_2 as functions of time and the phase portraits. Calculate the positions of the center of masses for both pendulum limbs and the pivot points over time, using the provided geometric relations.
3. **Energy Analysis.** Calculate the kinetic energy for each pendulum based on the angular velocities, masses, and moments of inertia. Compute the potential energy for each pendulum limb based on its *height*. Display the evolution of the energy with time. Check conservation of total energy, and discuss any deviations from conservation that you may find.
4. **Comparison with Experimental Data.** Ensure the numerical model uses the initial conditions derived from the experimental session for an accurate comparison. Plot and compare the experimental and numerical results for the angular displacements θ_1 and θ_2 over time, and the trajectories of the pivot points and center of masses. Discuss any discrepancies and analyze potential sources of error.
5. **Perturbation in initial conditions (Extra Credit).** Using the numerical model, address the effect of a small perturbation in the initial conditions. Consider scenarios with $\theta_1(t=0) = \theta_2(t=0) = \theta_0$ and $\dot{\theta}_1(t=0) = \dot{\theta}_2(t=0) = 0$. Simulate the case with $\theta_0 = 45^\circ$ and compare the results against a simulation with $\theta_0 = 45^\circ + \varepsilon$ (where $\varepsilon \ll 1$, you can take $\varepsilon = 10^{-10}$ rad). Repeat the same procedure for $\theta_0 = 135^\circ$ and $\theta_0 = 135^\circ + \varepsilon$. Discuss your findings and comment on the predictability of the motion of a double pendulum.

3.4 Mathematical model for the rigid-body compound double pendulum

Assuming an even mass distribution for the limbs implies that the centers of mass lie along the plane of symmetry for each limb, i.e. aligning the vector $\mathbf{P}_1\mathbf{P}_2$ (from pivot 1 to pivot 2) with the vector $\mathbf{P}_1\mathbf{G}_1$ (from pivot 1 to the center of mass of pendulum 1). This configuration often represents a more symmetric and physically intuitive setup, making the analysis and numerical integration more straightforward.

3.4.1 Kinematics

Based on the geometry of the double pendulum and the sketch in the figure 3.1, we can extract the following relationships from the geometry of the double pendulum, without using any information about forces:

$$\begin{aligned}\mathbf{P}_1\mathbf{G}_1 &= \mathbf{r}_0^{G_1} = R_1 \sin(\theta_1)\mathbf{i} - R_1 \cos(\theta_1)\mathbf{j} \\ \mathbf{P}_1\mathbf{P}_2 &= \mathbf{r}_0^{P_2} = L_1 \sin(\theta_1)\mathbf{i} - L_1 \cos(\theta_1)\mathbf{j} \\ \mathbf{P}_2\mathbf{G}_2 &= (\mathbf{r}_0^{G_2} - \mathbf{r}_0^{P_2}) = R_2 \sin(\theta_2)\mathbf{i} - R_2 \cos(\theta_2)\mathbf{j}\end{aligned}\tag{3.1}$$

This geometric relationships are used to extract the position of the center of mass for the first pendulum

$$\begin{aligned}\mathbf{r}_0^{G_1} &= \mathbf{P}_1\mathbf{G}_1 \\ x_1 &= R_1 \sin(\theta_1) \\ y_1 &= -R_1 \cos(\theta_1)\end{aligned}\tag{3.2}$$

and for the second pendulum

$$\begin{aligned}\mathbf{r}_0^{G_2} &= \mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{G}_2 \\ x_2 &= L_1 \sin(\theta_1) + R_2 \sin(\theta_2) \\ y_2 &= -L_1 \cos(\theta_1) - R_2 \cos(\theta_2)\end{aligned}\tag{3.3}$$

Taking the first derivative with respect to time to obtain the velocity:

$$\begin{aligned}\mathbf{v}_0^{G_1} &= \dot{\mathbf{r}}_0^{G_1} \\ \dot{x}_1 &= \dot{\theta}_1 R_1 \cos(\theta_1) \\ \dot{y}_1 &= \dot{\theta}_1 R_1 \sin(\theta_1)\end{aligned}\tag{3.4}$$

$$\begin{aligned}\mathbf{v}_0^{G_2} &= \dot{\mathbf{r}}_0^{G_2} \\ \dot{x}_2 &= \dot{\theta}_1 L_1 \cos(\theta_1) + \dot{\theta}_2 R_2 \cos(\theta_2) \\ \dot{y}_2 &= \dot{\theta}_1 L_1 \sin(\theta_1) + \dot{\theta}_2 R_2 \sin(\theta_2)\end{aligned}\tag{3.5}$$

Taking the second derivative with respect to time to obtain the acceleration:

$$\begin{aligned}\mathbf{a}_0^{G_1} &= \ddot{\mathbf{r}}_0^{G_1} \\ \ddot{x}_1 &= -\dot{\theta}_1^2 R_1 \sin(\theta_1) + \ddot{\theta}_1 R_1 \cos(\theta_1) \\ \ddot{y}_1 &= \dot{\theta}_1^2 R_1 \cos(\theta_1) + \ddot{\theta}_1 R_1 \sin(\theta_1)\end{aligned}\tag{3.6}$$

$$\begin{aligned}\mathbf{a}_0^{G_2} &= \ddot{\mathbf{r}}_0^{G_2} \\ \ddot{x}_2 &= -\dot{\theta}_1^2 L_1 \sin(\theta_1) + \ddot{\theta}_1 L_1 \cos(\theta_1) - \dot{\theta}_2^2 R_2 \sin(\theta_2) + \ddot{\theta}_2 R_2 \cos(\theta_2) \\ \ddot{y}_2 &= \dot{\theta}_1^2 L_1 \cos(\theta_1) + \ddot{\theta}_1 L_1 \sin(\theta_1) + \dot{\theta}_2^2 R_2 \cos(\theta_2) + \ddot{\theta}_2 R_2 \sin(\theta_2)\end{aligned}\tag{3.7}$$

3.4.2 Forces

Let $\mathbf{T}_1 = T_{1x}\mathbf{i} + T_{1y}\mathbf{j}$ be the force vector acting on pendulum 1 at pivot 1. Let $\mathbf{T}_2 = T_{2x}\mathbf{i} + T_{2y}\mathbf{j}$ be the force vector acting on pendulum 1 at pivot 2. By Newton's third law, $-\mathbf{T}_2$ is the force vector acting on pendulum 2 at pivot 2. From Newton's laws of motion, we can write the following force and moment equations:

$$m_1 \mathbf{a}_0^{G_1} = \mathbf{T}_1 + \mathbf{T}_2 - m_1 g \mathbf{j}$$

$$m_1 \ddot{x}_1 = T_{1x} + T_{2x} \quad (3.8)$$

$$m_1 \ddot{y}_1 = T_{1y} + T_{2y} - m_1 g \quad (3.9)$$

$$\begin{aligned} I_1 \ddot{\theta}_1 &= (-\mathbf{r}_0^{G_1}) \times \mathbf{T}_1 + (\mathbf{r}_0^{P_2} - \mathbf{r}_0^{G_1}) \times \mathbf{T}_2 \\ &= -(R_1 \sin(\theta_1) T_{1y} + R_1 \cos(\theta_1) T_{1x}) \\ &\quad + (L_1 \sin(\theta_1) - R_1 \sin(\theta_1)) T_{2y} + (L_1 \cos(\theta_1) - R_1 \cos(\theta_1)) T_{2x} \end{aligned} \quad (3.10)$$

Similarly, for the second pendulum:

$$m_2 \mathbf{a}_0^{G_2} = -\mathbf{T}_2 - m_2 g \mathbf{j}$$

$$m_2 \ddot{x}_2 = -T_{2x} \quad (3.11)$$

$$m_2 \ddot{y}_2 = -T_{2y} - m_2 g \quad (3.12)$$

$$I_2 \ddot{\theta}_2 = -(\mathbf{r}_0^{G_2} - \mathbf{r}_0^{P_2}) \times (-\mathbf{T}_2) = R_2 \sin(\theta_2) T_{2y} + R_2 \cos(\theta_2) T_{2x} \quad (3.13)$$

3.4.3 Equations of Motion

Substitute the expressions 3.6 and 3.20 for the acceleration of the center of mass into the forces (3.8,3.9,3.11,3.12) and moments equations (3.21 and 3.13). This yields modified versions of equations 3.8 through 3.13 with six unknowns: T_{1x} , T_{2x} , T_{1y} , T_{2y} , $\ddot{\theta}_1$, and $\ddot{\theta}_2$. We then solve for $\ddot{\theta}_1$ and $\ddot{\theta}_2$ and eliminate the other four unknowns.

The resulting equations for $\ddot{\theta}_1$ and $\ddot{\theta}_2$ are:

$$\begin{aligned} \ddot{\theta}_1 &= \frac{1}{\mathcal{D}} \left[-(2gm_1 R_1 (I_2 + m_2 R_2^2) \sin(\theta_1) + \right. \\ &\quad + L_1 m_2 (g(2I_2 + m_2 R_2^2) \sin(\theta_1) + \\ &\quad + R_2 (gm_2 R_2 \sin(\theta_1 - 2\theta_2) + 2(\dot{\theta}_2^2 (I_2 + m_2 R_2^2) + \\ &\quad \left. + \dot{\theta}_1^2 L_1 m_2 R_2 \cos(\theta_1 - \theta_2)) \sin(\theta_1 - \theta_2))) \right] \end{aligned} \quad (3.14)$$

$$\begin{aligned} \ddot{\theta}_2 &= \frac{1}{\mathcal{D}} \left[m_2 R_2 (- (g(2I_1 + L_1^2 m_2 + 2m_1 R_1^2) \sin(\theta_2)) + \right. \\ &\quad + L_1 (gm_1 R_1 \sin(\theta_2) + 2\dot{\theta}_1^2 (I_1 + L_1^2 m_2 + m_1 R_1^2) \sin(\theta_1 - \theta_2) + \\ &\quad + \dot{\theta}_2^2 L_1 m_2 R_2 \sin(2(\theta_1 - \theta_2)) + \\ &\quad \left. gm_1 R_1 \sin(2\theta_1 - \theta_2) + gL_1 m_2 \sin(2\theta_1 - \theta_2))) \right] \end{aligned} \quad (3.15)$$

defining $\mathcal{D}$ as the denominator, which is a function of the unknowns:

$$\begin{aligned} \mathcal{D} &= 2I_2 L_1^2 m_2 + 2I_2 m_1 R_1^2 + L_1^2 m_2^2 R_2^2 + 2m_1 m_2 R_1^2 R_2^2 + \\ &\quad 2I_1 (I_2 + m_2 R_2^2) - L_1^2 m_2^2 R_2^2 \cos(2(\theta_1 - \theta_2)) \end{aligned} \quad (3.16)$$

3.5 Deliverables

One student in each team will upload all the results to AulaGlobal in a single zipped file named `Group_Li_S1_S2_S3.zip`, where Li stands for the Lab Group (LA, or LB) and S1, S2 and S3 are the first surnames of the three students (e.g., `Group_L1_Sanchez_Gonzalez_Garcia`). The zipped file must contain the following:

1. **Report (maximum 10 pages):** A comprehensive report in PDF format that combines the results and analysis for both sessions. The report must address all previously described tasks, including experimental data processing, numerical modelling, energy analysis, and comparisons between experimental and numerical results. Critical analysis of the results is mandatory. Ensure that your figures are meaningful and readable. Ensure that the title page includes the full names of all three students. Name the report file as: `Report_Li_S1_S2_S3.pdf`
2. **Code:** The MATLAB code used for the experimental data processing and numerical integration. This must be organized into a sub-folder named: `code_Li_S1_S2_S3`. Inside this folder, the following scripts are required:
 - (a) `main_exp.m`: The main script for post-processing the experimental data. This script must produce all the results related to the experimental session when executed.
 - (b) `main_num.m`: The main script for solving the equations of motion numerically. This script must produce all the results related to the numerical session when executed.
 - (c) You can add any additional Matlab file (function or script *.m, or data *.mat) that is required to run your analysis.

Both scripts must run independently and reproduce all results without any additional input from the user. Ensure that all code is tested thoroughly before submission. Any code that fails to execute or does not produce the required results will incur a penalty. Note that your code must be standalone, i.e. it should be able to work with the content you upload. The provided code by the professor should not be required in the final code version.

3.6 Annex: Generalization for G_1 out of plane of symmetry

Let ϕ be the angle from vector $\mathbf{P}_1\mathbf{G}_1$ to $\mathbf{P}_1\mathbf{P}_2$. Note that ϕ is a constant, representing an uneven distribution of mass across the plane of symmetry of the limb. See figure 3.2.

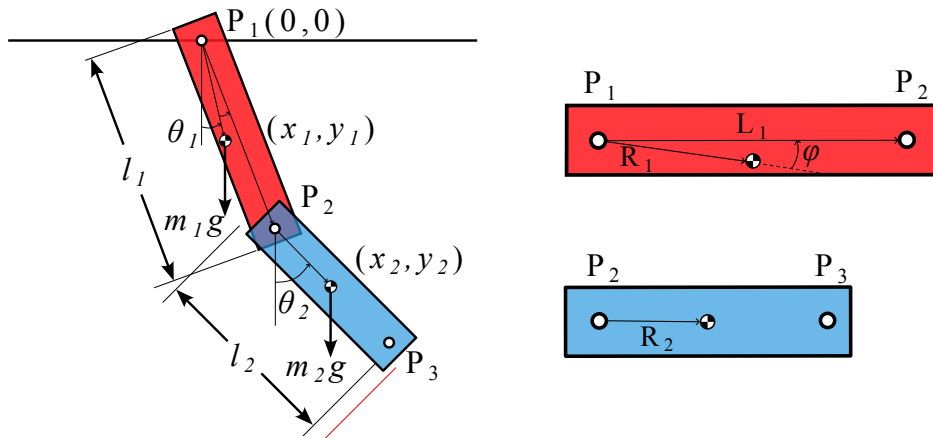


Figure 3.2: Rigid Body Double Pendulum Model for $\phi \neq 0$

3.6.1 Kinematics

Based on the geometry of the double pendulum and the sketch in the figure 3.2, we can extract the following relationships from the geometry of the double pendulum, without using any information about forces, which is slightly modified:

$$\mathbf{P}_1 \mathbf{P}_2 = \mathbf{r}_0^{P_2} = L_1 \sin(\theta_1 + \phi) \mathbf{i} - L_1 \cos(\theta_1 + \phi) \mathbf{j} \quad (3.17)$$

The kinematics of the first pendulum do not change; however, for the second pendulum we must account for $\phi \neq 0$

$$\begin{aligned} \mathbf{r}_0^{G_2} &= \mathbf{P}_1 \mathbf{P}_2 + \mathbf{P}_2 \mathbf{G}_2 \\ x_2 &= L_1 \sin(\theta_1 + \phi) + R_2 \sin(\theta_2) \\ y_2 &= -L_1 \cos(\theta_1 + \phi) - R_2 \cos(\theta_2) \end{aligned} \quad (3.18)$$

Taking the first derivative with respect to time to obtain the velocity:

$$\begin{aligned} \mathbf{v}_0^{G_2} &= \dot{\mathbf{r}}_0^{G_2} \\ \dot{x}_2 &= \dot{\theta}_1 L_1 \cos(\theta_1 + \phi) + \dot{\theta}_2 R_2 \cos(\theta_2) \\ \dot{y}_2 &= -\dot{\theta}_1 L_1 \sin(\theta_1 + \phi) + \dot{\theta}_2 R_2 \sin(\theta_2) \end{aligned} \quad (3.19)$$

Taking the second derivative with respect to time to obtain the acceleration:

$$\begin{aligned} \mathbf{a}_0^{G_2} &= \ddot{\mathbf{r}}_0^{G_2} \\ \ddot{x}_2 &= -\dot{\theta}_1^2 L_1 \sin(\theta_1 + \phi) + \ddot{\theta}_1 L_1 \cos(\theta_1 + \phi) - \dot{\theta}_2^2 R_2 \sin(\theta_2) + \ddot{\theta}_2 R_2 \cos(\theta_2) \\ \ddot{y}_2 &= \dot{\theta}_1^2 L_1 \cos(\theta_1 + \phi) + \ddot{\theta}_1 L_1 \sin(\theta_1 + \phi) + \dot{\theta}_2^2 R_2 \cos(\theta_2) + \ddot{\theta}_2 R_2 \sin(\theta_2) \end{aligned} \quad (3.20)$$

Forces and Moments.

No change in forces but we need to adjust the equation for the moments in the first pendulum:

$$\begin{aligned} I_1 \ddot{\theta}_1 &= (-\mathbf{r}_0^{G_1}) \times \mathbf{T}_1 + (\mathbf{r}_0^{P_2} - \mathbf{r}_0^{G_1}) \times \mathbf{T}_2 \\ &= -(R_1 \sin(\theta_1) T_{1y} + R_1 \cos(\theta_1) T_{1x}) \\ &\quad + (L_1 \sin(\theta_1 + \phi) - R_1 \sin(\theta_1)) T_{2y} + (L_1 \cos(\theta_1 + \phi) - R_1 \cos(\theta_1)) T_{2x} \end{aligned} \quad (3.21)$$

3.6.2 Equations of Motion

The resulting equations for $\ddot{\theta}_1$ and $\ddot{\theta}_2$ are:

$$\begin{aligned} \ddot{\theta}_1 &= \frac{1}{\mathcal{D}} \left[-(2gm_1 R_1 (I_2 + m_2 R_2^2) \sin(\theta_1) + \right. \\ &\quad + L_1 m_2 (g(2I_2 + m_2 R_2^2) \sin(\theta_1 + \phi) + \\ &\quad + R_2 (gm_2 R_2 \sin(\theta_1 - 2\theta_2 + \phi) + 2(\dot{\theta}_2^2 (I_2 + m_2 R_2^2) + \\ &\quad \left. + \dot{\theta}_1^2 L_1 m_2 R_2 \cos(\theta_1 - \theta_2 + \phi)) \sin(\theta_1 - \theta_2 + \phi))) \right] \end{aligned} \quad (3.22)$$

$$\begin{aligned} \ddot{\theta}_2 &= \frac{1}{\mathcal{D}} \left[m_2 R_2 (- (g(2I_1 + L_1^2 m_2 + 2m_1 R_1^2) \sin(\theta_2)) + \right. \\ &\quad + L_1 (gm_1 R_1 \sin(\theta_2 - \phi) + 2\dot{\theta}_1^2 (I_1 + L_1^2 m_2 + m_1 R_1^2) \sin(\theta_1 - \theta_2 + \phi) + \\ &\quad + \dot{\theta}_2^2 L_1 m_2 R_2 \sin(2(\theta_1 - \theta_2 + \phi)) + \\ &\quad \left. gm_1 R_1 \sin(2\theta_1 - \theta_2 + \phi) + gL_1 m_2 \sin(2\theta_1 - \theta_2 + 2\phi))) \right] \end{aligned} \quad (3.23)$$

defining $\mathcal{D}$ as the denominator, which is a function of the unknowns:

$$\begin{aligned} \mathcal{D} &= 2I_2 L_1^2 m_2 + 2I_2 m_1 R_1^2 + L_1^2 m_2^2 R_2^2 + 2m_1 m_2 R_1^2 R_2^2 + \\ &\quad 2I_1 (I_2 + m_2 R_2^2) - L_1^2 m_2^2 R_2^2 \cos(2(\theta_1 - \theta_2 + \phi)) \end{aligned} \quad (3.24)$$